Supplementary Material:
Ceará Digital Twin for Wildfires

Gois et al.

1 Exp-B iteration tables (v1–v9)

Full Exp-B tables are reproduced from `figures/experimentos-artigo.tex` in the main manuscript. This supplement archives machine-readable logs under `backend/experiments/results/`.

2 Bootstrap and paired significance tests

Table 1: Bootstrap 95% CI for published Exp-B v9 alert metrics (2,000 resamples on contingency tables).

Metric	**Point**	**CI low**	**CI high**
XGBoost YES precision (23 TP, 5 FP)	82.1%	67.9%	96.4%
NeKo-PIGNN YES precision (11 TP, 1 FP)	91.7%	75.0%	100.0%
Combined coverage (YES+UNCERTAIN)	88.0%	84.6%	91.2%
MLP RMSE (5 seeds, reproduced)	—	see Table ??	—

Table 2: Paired significance tests: Wilcoxon signed-rank (regression RMSE) and exact McNemar (classification).

Test	**Statistic**	***p*-value**	**Sig. @0.05**	
Wilcoxon: MLP vs XGB RMSE	0.00	0.0625	No	
McNemar: 3-class XGB vs persistence	15	0.1418	No	acc 73.2McNemar: YES alert X
2	0.8714	No	prec 100.0	

Table 3: Extended vs. short-window INPE benchmarks (XGBoost 3-class, YES at $P \geq 0.30$).

Window	Days	Focos	n	Prec.	Rec.	FP
Short (Mar–Jun/26)	117	866	340	34.4%	38.9%	40
Dry season 2024	141	2331	412	44.9%	66.4%	114
Dry season 2025	184	24573	541	84.2%	91.8%	75
Full 12 mo	316	3789	936	31.6%	74.1%	130
LOO: tr24/te25	500	27984	2745	79.0%	85.9%	379
LOO: tr25/te24	500	27984	2070	39.2%	98.2%	595

Table 4: External-state validation: XGBoost 3-class YES alert at $P \geq 0.30$ (dry season Jul–Dec/2025).

State	Days	Hotspots	Precision	Recall
Ceará	184	24573	84.2%	91.8%
Maranhão	184	233854	57.5%	99.5%
Piauí	183	145220	60.1%	96.4%

Table 5: Operational baseline comparison: NASA FIRMS, INPE BDQueimadas, and LangGraph pipeline.

System	Latency	Prec.	Rec.	Cost
NASA FIRMS direct	3–6 h	80.0%	15.7%	Free
INPE BDQueimadas	3–6 h	—	—	Free
LangGraph (ours)	52 s / <1 s	84.2%	91.8%	$20

FIRMS→INPE spatial match (1 km, same day): 25.8% of 128 FIRMS points; 8-day sample.

Table 6: Formal RAG retrieval benchmark (50 questions, bilingual TF-IDF proxy of FAISS pipeline).

Metric	Value
Questions	50
Indexed chunks	32
Recall@5	98.0%
MRR	0.907

3 Extended temporal and external-state validation

4 Operational baseline comparison

5 RAG benchmark

6 Model hyperparameters

Table 7: NeKo-PIGNN and alert-model hyperparameters.

Component	**Setting**
KoopmanAE encoder	MLP 512-256-128, $d = 64$, Adam 10^{-4}, batch 32
PI-GNN	3×GCNConv, 128 channels, $r = 3$ cells, $\lambda_2 = 0.05$
XGBoost 3-class	300–400 trees, depth 4–5, lr 0.04–0.05
YES alert threshold	$P(\text{YES}) \geq 0.30$
RAG index	chunk 900, overlap 120, top_k=5, BGE-small-en-v1.5

7 Experiment JSON logs

Reproduce with:

```
cd backend && python -m experiments.statistical_robustness
cd backend && python -m experiments.extended_temporal_benchmark
cd backend && python -m experiments.external_state_benchmark
cd backend && python -m experiments.operational_baseline_benchmark
cd backend && python -m experiments.rag_benchmark
```